

Highlights

TrustFedGNN: Contribution-Based Aggregation without Self-Reported Metadata for Trustworthy Federated Graph Learning

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- Fairness aggregation that reads no client-reported metadata closes an attack surface
- A falsifying client moves our weights by 0.0000; it takes 86% of a Frank–Wolfe one
- Statistic-level DP costs $O(1)$ noise and 0.002 AUC, against 0.202 for full DP-SGD
- Best AUC over 7 federated baselines on Pokec-z ($p=0.0020$, Holm, 10/10 seeds)
- Four algebraic guarantees machine-checked in Lean 4; seven refuted hypotheses listed

TrustFedGNN: Contribution-Based Aggregation without Self-Reported Metadata for Trustworthy Federated Graph Learning

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Abstract

Cross-silo federated graph learning for credit scoring, bail assessment and anti-money-laundering must satisfy three requirements simultaneously: demographic fairness, differential privacy, and resilience to malicious participants. Every deployed fairness-aware federated aggregator we are aware of steers on disparity statistics that clients report about themselves, which turns fairness steering into an attack surface: a participant that reports zero disparity while uploading a poisoned update can capture the aggregate. TrustFedGNN closes that surface. On the client, Fairness-Sensitive Edge Reweighting attenuates cross-group attention, and Fairness-Targeted Gradient Decomposition privatises only the two-dimensional group-mean statistic, obtaining (ϵ, δ) -differential privacy at $O(1)$ noise rather than the $O(\sqrt{P})$ of full-parameter DP-SGD. On the server, FU-Shapley scores each client by the alignment of its update with a target gradient the server computes on its own holdout split, then gates those scores onto the probability simplex; no self-reported field enters the weights. We prove metadata immunity and confirm it empirically: a falsifying client leaves the weight vector bit-identical ($\|\Delta \mathbf{w}\|_\infty = 0.0000$) while capturing 86.2% of a Frank–Wolfe aggregate under the identical attack. Four algebraic properties are machine-checked in Lean 4. Across four demographic benchmarks and two large-scale graphs, TrustFedGNN attains the best AUC among seven external baselines

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on Pokec-z ($p = 0.0020$ after Holm–Bonferroni correction, winning on all ten seeds), and is statistically indistinguishable from all of them on a synthetic k -nearest-neighbour tabular graph. We report seven pre-registered hypotheses that our own measurements refuted.

Keywords: Federated Learning, Graph Neural Networks, Algorithmic Fairness, Differential Privacy, Byzantine Robustness, Contribution Valuation

1. Introduction

Graph neural networks (GNNs) are now the default model class for fraud detection, anti-money-laundering (AML) and credit assessment [1, 2, 3], because the entities in those domains are defined as much by their relations—transactions, shared devices, social ties—as by their own attributes. In cross-institutional settings, however, data-protection regimes such as the EU General Data Protection Regulation forbid pooling those graphs into one repository. The resulting setting is *cross-silo horizontal federated graph learning*: K institutions each hold an induced subgraph, inter-silo edges are severed, and only model updates cross the boundary [4, 5].

Deploying such a system in a high-stakes domain requires three properties at once, and each pair of them is in tension.

Structural bias against group fairness.. Nodes sharing a sensitive attribute are densely interconnected—demographic homophily—so recursive neighbourhood aggregation propagates and amplifies historical bias [6, 7].

Group fairness against differential privacy.. Enforcing a fairness constraint requires the server to observe some disparity signal, and that signal leaks group membership. The textbook remedy, DP-SGD over the full parameter vector [8], injects noise scaling as $O(\sqrt{P})$ and destroys GNN utility: in our own ablation it costs 0.2016 AUC (Section 7.3).

Fairness steering against Byzantine resilience.. This is the tension we consider unresolved, and it motivates the paper. Fairness-aware federated aggregators—FairFed [9], PUFFLE [10], F²GNN [11], FairGFL [12]—compute aggregation weights from disparity statistics *that each client reports about itself*. Nothing in those protocols verifies the report. An adversary may therefore transmit a poisoned update while declaring $\widehat{\text{DPD}}_k = 0$, and the fairness mechanism

will reward it precisely because it claims to be fair. Kasyap et al. [13] and EAB-FL [14] demonstrate this class of attack; classical Byzantine defences [15, 16, 17] do not address it, because they screen the parameter update and never examine the metadata channel at all.

1.1. Contributions

TrustFedGNN removes the metadata channel rather than trying to secure it. Its server computes the reference signal it needs *itself*, on a holdout split no client can touch, and derives aggregation weights from how well each client’s update aligns with that reference. A client’s report about its own fairness is never read, so falsifying it is mathematically inert.

- **Pillar C1 — privacy-aware fairness feedback at the client (Section 4.1).** Fairness-Sensitive Edge Reweighting (FSER) attenuates attention on cross-group edges before the softmax. Fairness-Targeted Gradient Decomposition (FTGD) projects the update orthogonally to the fairness-ascent direction and injects Gaussian noise into the released two-dimensional statistic (μ_0, μ_1) only, computed on a sensitive-blind forward pass. This yields a certified (ϵ, δ) -DP release at $O(1)$ noise, at a measured utility cost of 0.0017 AUC against 0.2016 for full-parameter DP-SGD—a factor of 115.
- **Pillar C2 — metadata-independent contribution aggregation at the server (Section 4.2).** FU-Shapley scores client k by $\varphi_k = \langle g_k, g_{\text{target}} \rangle / \|g_{\text{target}}\|$, where $g_{\text{target}} = g_{\text{task}}^{\text{srv}} + \alpha g_{\text{fair}}^{\text{srv}}$ is computed on the server holdout; scores are smoothed by an exponential moving average and gated onto the simplex by a rectifier. Theorem 4 establishes that the weights are a function of the transmitted updates alone. Empirically, a client that falsifies its report leaves the weight vector bit-identical, while the same falsification hands it 86.2% of a Frank–Wolfe aggregate (Table 10).
- **Pillar C3 — evidence, with its boundaries stated (Sections 7–8).** Four demographic-fairness benchmarks and two large-scale graphs using proxy groups; seventeen implemented baselines; ten seeds with paired Wilcoxon tests under Holm–Bonferroni correction; four algebraic properties machine-checked in Lean 4. We also report seven pre-registered hypotheses that our measurements refuted, including the utility attribution of FSER itself.

Our central claim is deliberately bounded, and we state it once, in the form we can defend against every table in this paper:

Among the methods surveyed here, TrustFedGNN is the only one that holds all four trust properties simultaneously at an acceptable level, and the only one whose fairness-aware aggregation does not depend on client-reported metadata. On individual axes, specialised methods still score better, and we report every such case.

2. Related Work

We organise the literature along the five axes that the problem forces together.

2.1. Fair graph neural networks

Centralised fair-GNN methods intervene at the representation, the input, or the neighbourhood. FairGNN [6] trains an adversarial discriminator to purge sensitive information from embeddings under partial attribute observation; NIFTY [7] adds counterfactual and Lipschitz-stability regularisation; EDITS [18] learns a debiasing transformation of features and adjacency before message passing; FairSIN [19] neutralises sensitive information by injecting fairness-aware auxiliary features. More recent work recasts fairness as class-and-group rebalancing [20], seeks representations invariant across unseen sensitive attributes [21], disentangles then selectively debiases latent factors [22], or regularises the divergence between group score distributions rather than their means [23]. All assume centralised access to the full graph.

2.2. Fairness in federated learning

Two distinct notions travel under the name. *Performance uniformity* equalises accuracy across clients: q -FedAvg [24] reweights toward high-loss participants. *Group fairness* equalises outcomes across sensitive-attribute groups irrespective of client boundaries; this is the notion we adopt. In that second family, FairFed [9] adjusts aggregation weights from client-reported disparity gaps, PUFFLE [10] co-tunes a privacy budget and a fairness coefficient, FedFACT [25] and FDP-Fair [26, 27] post-process or dualise the constraint, and the PoPETs-track framework of Bendoukha et al. [28] combines secure aggregation with fairness-aware selection. F²GNN [11] is the

closest structure-aware federated attempt, and FairGFL [12] and FedGraph-Fair [29] are recent federated graph systems evaluated on financial data; both appear as baselines in Section 7. The common structural weakness is stated above: the fairness signal is self-reported and unverified.

2.3. Differential privacy and Byzantine robustness

DP-SGD [8, 30, 31] clips and perturbs the full update, accounted through Rényi DP [32] and converted to (ϵ, δ) by the analytic formula of Canonne, Kamath and Steinke [33]. Because $P \geq 10^5$ for even compact GNNs, isotropic noise is ruinous—quantified as arm M4 in Section 7.2. Byzantine-robust rules screen updates geometrically: Krum [15], coordinate-wise median and trimmed mean [16], FLTrust [17]. A substantial literature shows these can be evaded by adversaries that stay inside the benign geometric envelope [34, 35, 36, 37]. None of it considers an adversary whose parameter update is unremarkable but whose *auxiliary fairness metadata* is fabricated—the threat we formalise in Section 3.2.

2.4. Contribution valuation

Our aggregator descends from the contribution-valuation literature rather than from constrained optimisation. GTG-Shapley [38] estimates a client’s marginal contribution as a cooperative-game value; CGSV [39] scores by cosine alignment between client and aggregate gradients; further schemes translate contribution into differentiated rewards [40, 41, 42, 43]. TrustFedGNN’s departure is the reference against which alignment is measured: CGSV aligns against the aggregate of the very updates being scored, which an adversarial majority can shift, whereas we align against a target gradient computed on data the server holds privately, and we make that target explicitly bi-objective so that fairness contribution is scored, not merely utility. Exact Shapley computation is $O(2^K)$ and infeasible here; we return to what our $O(KP)$ surrogate does and does not inherit from it in Section 7.7.

2.5. Positioning

Table 1 places TrustFedGNN against the closest work on four axes: whether the method operates on graphs, whether it targets group-demographic fairness, whether its privacy guarantee is certified rather than heuristic, and—the axis that motivates this paper—whether its aggregation is independent of client-reported metadata.

Table 1: **Positioning against closely related frameworks.** “Metadata-independent” means the aggregation weights are provably a function of transmitted updates only, with no self-reported client field entering the computation (Theorem 4). “Certified privacy” means a formal (ϵ, δ) - or Rényi-DP guarantee rather than an empirical privacy claim; † marks a guarantee scoped to the released fairness statistic rather than to the full update (Section 8).

Framework	Graph-aware	Group fairness	Certified privacy	Metadata-independent
FairGNN [6]	✓	✓ (centralised)	×	—
FairSIN [19]	✓	✓ (centralised)	×	—
FairFed [9]	×	✓ (heuristic)	×	×
PUFFLE [10]	×	✓	✓ (full DP-SGD)	×
F ² GNN [11]	✓	✓	×	×
FairGFL [12]	✓	✓ (overlap)	✓ (local DP)	×
FedGraph-Fair [29]	✓	✓ (DRO)	×	×
CGSV [39]	×	×	×	✓
Coordinate median [16]	×	×	×	✓
BFWA (baseline, Section 4.3)	×	✓ (dual $\leq \tau$)	×	×
TrustFedGNN (ours)	✓	✓ ‡	✓ †	✓

‡ The fairness column records a targeted, utility-preserving steering mechanism. We do not claim superiority on every disparity metric; Section 8 reports the cases where a baseline is fairer than we are.

3. Problem Formulation and Threat Model

3.1. Cross-silo federated graph learning

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ be partitioned across K institutions. Client k holds $\mathcal{G}_k = (\mathcal{V}_k, \mathcal{E}_k, \mathbf{X}_k)$ with features $\mathbf{X}_k \in \mathbb{R}^{n_k \times d}$, binary labels $\mathbf{y}_k$, and binary sensitive attributes $\mathbf{s}_k$; inter-silo edges are severed and cannot be exchanged. The federation minimises

$$\min_{\theta \in \mathbb{R}^P} F(\theta) = \sum_{k=1}^K \frac{n_k}{N} \mathcal{L}_{\text{task}}^{(k)}(\theta) \quad \text{s.t.} \quad \text{DPD}(\theta) \leq \tau, \quad (1)$$

with $\text{DPD}(\theta) = |\mathbb{E}[\hat{y}_v \mid s_v = 1] - \mathbb{E}[\hat{y}_v \mid s_v = 0]|$ and τ an operator-set tolerance. We report the *hard* disparity DPD_{hard} , computed on thresholded labels, in all tables; the differentiable surrogate DPD_{soft} on probability scores is used only inside the loss. The two are never mixed.

3.2. Threat model

Two adversaries operate simultaneously.

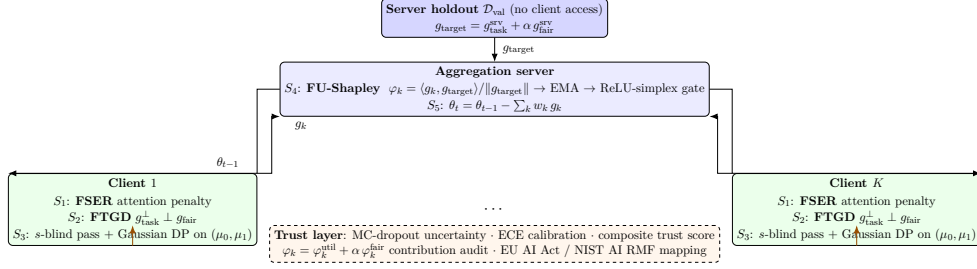


Figure 1: **The closed $S_0 \rightarrow S_5$ loop.** Clients upload only the pseudo-gradient g_k ; the privatised disparity $\widehat{\text{DPD}}_k$ accompanies it for logging and audit but—this is the point of the design—never enters the weight computation. The reference signal g_{target} is manufactured by the server on a holdout split that no client can reach, which is what makes a falsified client report inert.

Honest-but-curious server.. It follows the protocol but attempts to infer $\mathbf{s}_k$ from everything it receives: the released fairness statistic and the raw parameter update $g_k = \theta_t - \theta_k$. We treat these as two distinct channels throughout, because our guarantee covers one and not the other.

Adaptive stealth fairness poisoner.. Up to $f < K/2$ clients are malicious and omniscient. Such a client constructs a poisoning update g_{poison} that maximises disparity, then projects it into a safe ball around the benign median so that geometric screens do not fire,

$$g_{\text{stealth}} = \text{med}(G) + \min\left(1, \frac{r_{\text{safe}}}{\|g_{\text{poison}} - \text{med}(G)\|_2}\right)(g_{\text{poison}} - \text{med}(G)), \quad (2)$$

with $r_{\text{safe}} = 0.85 \cdot R_{\text{med}}$, and *simultaneously falsifies its metadata*, declaring $\widehat{\text{DPD}}_k = 0.0$ and $\text{Perf}_k = 0.99$. The falsification is the operative part: it is what turns a fairness-steering aggregator from a defence into a lever.

4. TrustFedGNN

One communication round proceeds through six stages, illustrated in Fig. 1 and consolidated in Algorithm 1.

4.1. Client-side mechanisms (Pillar C1)

4.1.1. S_1 : Fairness-Sensitive Edge Reweighting

At GAT layer l , attention logits are attenuated on edges that cross the sensitive boundary:

$$\tilde{e}_{vu}^{(l)} = e_{vu}^{(l)} - \beta^{(l)} \cdot \mathbb{I}(s_v \neq s_u) \cdot \max\left(0, \cos(\mathbf{h}_v^{(l-1)}, \mathbf{h}_u^{(l-1)})\right), \quad (3)$$

with $\beta^{(l)} \geq 0$ learned and clamped to $[0, 5]$ to keep the softmax numerically well-posed. The penalty fires only where two conditions hold at once: the edge crosses groups, and the endpoint representations are already similar. Its intended role is to damp the specific pathway by which homophily converts into predictive disparity, while leaving task-relevant homophilic structure untouched.

Equation (3) has a property worth stating because it determines where the mechanism can possibly act. If a graph contains no cross-group edges, the indicator is identically zero and FSER reduces exactly to plain GAT attention—a mathematical no-op, not a weak effect. This is why the Elliptic Bitcoin graph, whose temporal proxy groups yield $h_s = 1.0000$ by construction, cannot serve as a measurement point for FSER at all (Section 6.1).

We do not claim a causal separation between demographic similarity and behavioural similarity; FSER is a learned heuristic, and Section 7.2 reports that its measured contribution to utility on our benchmarks is null.

4.1.2. S_2 : Fairness-Targeted Gradient Decomposition

Let $g_{\text{total}} = \nabla_{\theta}(\mathcal{L}_{\text{task}} + \lambda\mathcal{L}_{\text{fair}})$ and $g_{\text{fair}} = \nabla_{\theta}(\lambda\mathcal{L}_{\text{fair}})$. FTGD removes from the step any component along the fairness-ascent direction:

$$g_{\text{task}}^{\perp} = g_{\text{total}} - \frac{\langle g_{\text{total}}, g_{\text{fair}} \rangle}{\|g_{\text{fair}}\|^2 + \varepsilon} g_{\text{fair}}. \quad (4)$$

The projection is guarded by a minimum-norm threshold on $\|g_{\text{fair}}\|$ rather than an arbitrary ε , so a vanishing fairness gradient near convergence does not amplify numerical noise.

4.1.3. S_3 : Statistic-level differential privacy

The sensitive attribute enters the objective through exactly two scalars, the group prediction means $\mu_g = |\mathcal{V}_{k,g}|^{-1} \sum_{v \in \mathcal{V}_{k,g}} \hat{y}_v$. The task loss does not depend on $\mathbf{s}_k$ at all. This is the observation the mechanism rests on: privatise the two scalars, not the P -dimensional parameter vector,

$$\tilde{\mu}_g = \mu_g + \mathcal{N}(0, \sigma_{\text{DP}}^2), \quad \widetilde{\text{DPD}}_k = |\tilde{\mu}_1 - \tilde{\mu}_0|. \quad (5)$$

One subtlety is load-bearing. Because FSER consumes $\mathbf{s}$ inside the attention mechanism, the network’s prediction for node v depends on the sensitive

attributes of v 's whole L -hop neighbourhood, not on s_v alone—which would invalidate the per-node sensitivity bound of Assumption 4. TrustFedGNN therefore computes the released statistic on a second, *sensitive-blind* forward pass, in which the FSER penalty is disabled, while training continues to use the real $\mathbf{s}$. The cost is one extra forward pass per round; the benefit is that Proposition 1 applies to the quantity actually released.

4.2. Server-side aggregation (Pillar C2)

4.2.1. S_4 : FU-Shapley

The server maintains a holdout split $\mathcal{D}_{\text{val}}$ that no client contributes to or observes. On it, it computes its own bi-objective reference direction

$$g_{\text{target}} = g_{\text{task}}^{\text{srv}} + \alpha g_{\text{fair}}^{\text{srv}}, \quad g_{\text{fair}}^{\text{srv}} = \nabla_{\theta} (\mu_0^{\text{val}} - \mu_1^{\text{val}})^2, \quad (6)$$

with $\alpha = 0.1$. Three details of Eq. (6) are easy to get wrong and change the mechanism's meaning. The sign is *positive*: $g_{\text{fair}}^{\text{srv}}$ ascends the disparity penalty, so a client that reduces disparity aligns positively with it and earns positive fairness credit; a minus sign would penalise exactly the clients one wants to reward. The server-side surrogate is the *squared* gap, which is smooth and whose gradient attenuates as fairness is achieved, whereas clients retain the non-smooth $|\mu_0 - \mu_1|$; the asymmetry is deliberate. Finally, coordinates are laid out following `state_dict()` rather than `parameters()`, so that the scoring inner product is aligned coordinate-for-coordinate with the flattened pseudo-gradients, buffers without gradients being zero-filled.

Client k 's instantaneous contribution and its round-normalised form are

$$\varphi_k^{\text{raw}} = \langle g_k, g_{\text{target}} \rangle, \quad \varphi_k = \frac{\varphi_k^{\text{raw}}}{\|g_{\text{target}}\| + 10^{-8}}. \quad (7)$$

Normalising by $\|g_{\text{target}}\|$ removes the scale drift that would otherwise occur as $g_{\text{task}}^{\text{srv}} \rightarrow 0$ near convergence, and—unlike a z -score, which re-centres and therefore gates roughly half the clients at every round regardless of behaviour—it preserves the sign, so that the rectifier below retains its intended meaning of “actively harmful” rather than “below average”. Scores are smoothed across rounds,

$$\bar{\varphi}_k^{(t)} = \beta_{\text{ema}} \bar{\varphi}_k^{(t-1)} + (1 - \beta_{\text{ema}}) \varphi_k^{(t)}, \quad \beta_{\text{ema}} = 0.9, \quad (8)$$

and gated onto the simplex with an explicit null-player mask,

$$w_k = \frac{\max(0, \bar{\varphi}_k) \cdot \mathbb{I}(g_k \neq \mathbf{0})}{\sum_{j=1}^K \max(0, \bar{\varphi}_j) \cdot \mathbb{I}(g_j \neq \mathbf{0})}. \quad (9)$$

Finiteness contract.. A non-finite score is an absorbing state of the recursion in Eq. (8), and it has a security consequence that is not obvious. Under a sign-flip attack the adversary’s gradient can explode, making φ non-finite; since `nan` $>$ 0 evaluates to false, a naive implementation of Eq. (9) would silently degenerate to uniform averaging and hand the attacker its full $1/K$ share. TrustFedGNN therefore retains the previous EMA value for any coordinate whose score is non-finite, and zeroes it at the first round—an undefined contribution earns no credit. When *every* score is non-positive the rule falls back to uniform weighting over active clients, and the two causes are logged distinctly: `degenerate_nan` (the measurement failed) versus `degenerate_nonpos` (all clients genuinely harmed the objective).

Robust variant.. `robust_fu_shapley` prepends a screening step, discarding the f updates furthest from the coordinate-wise median before scoring the survivors. Section 7.6 shows this is not redundant: the rectifier alone suffices against sign-flip and fairness poisoning but is defeated by gradient scaling, which the screen closes.

Explainability by construction.. Bilinearity of the inner product splits each score into an audit trail without any additional computation, $\varphi_k = \varphi_k^{\text{util}} + \alpha \varphi_k^{\text{fair}}$ (Proposition 7). A positive φ_k^{fair} means client k moved the global model toward parity. The federation operator therefore obtains a per-client, per-round statement of *why* a participant was weighted as it was—a governance artifact rather than a post-hoc rationalisation.

4.2.2. S_5 : Global update and trust layer

The server applies $\theta_{t+1} = \theta_t - \eta \sum_k w_k g_k$ and measures MC-dropout predictive variance [44], expected calibration error, a composite trust score, and the aggregation-weight total variation $\Omega_w = \sum_t \|\mathbf{w}^{(t)} - \mathbf{w}^{(t-1)}\|_1$.

Remark 1 (Reading Ω_w). $\Omega_w > 0$ is a design consequence, not instability. FedAvg attains $\Omega_w = 0$ because its weights are fixed sample proportions and carry no information about behaviour; any rule that re-scores participants each round must move. The meaningful comparison is against another adaptive rule, and there FU-Shapley is $27\times$ steadier than the Frank–Wolfe baseline (Table 19).

4.3. The BFWA baseline

For comparison we also implement Bi-objective Frank–Wolfe Aggregation (BFWA), which treats Eq. (1) as a constrained programme over the simplex,

Algorithm 1 TrustFedGNN: one communication round

Require: Global parameters θ_t ; server holdout $\mathcal{D}_{\text{val}}$; fairness weight α ; EMA factor β_{ema} ; DP scale σ_{DP} ; EMA state $\bar{\varphi}^{(t-1)}$

Ensure: Updated parameters θ_{t+1} , updated EMA state

- 1: Server broadcasts θ_t to all clients $\{S_0\}$
 - 2: **for** each client k **in parallel** **do**
 - 3: Train E local epochs with FSER attention, Eq. (3) $\{S_1\}$
 - 4: At each step, take the FTGD direction $g_{\text{task}}^\perp$, Eq. (4) $\{S_2\}$
 - 5: Recompute (μ_0, μ_1) on an s -blind pass; release $\tilde{\mu}_g$, Eq. (5) $\{S_3\}$
 - 6: Upload $g_k = \theta_t - \theta_k$, and $\widetilde{\text{DPD}}_k$ *for logging only*
 - 7: **end for**
 - 8: Server computes g_{target} on $\mathcal{D}_{\text{val}}$, Eq. (6) $\{S_4\}$
 - 9: (*robust variant*) discard the f updates furthest from $\text{med}(\{g_k\})$
 - 10: Score φ_k by Eq. (7); smooth by Eq. (8); gate by Eq. (9)
 - 11: $\theta_{t+1} \leftarrow \theta_t - \eta \sum_k w_k g_k$ $\{S_5\}$
 - 12: **return** $\theta_{t+1}, \bar{\varphi}^{(t)}$
-

$\max_{w \in \Delta_K} \sum_k w_k \text{Perf}_k$ subject to $\sum_k w_k \widetilde{\text{DPD}}_k \leq \tau$, solved by Frank–Wolfe steps with a persistent dual multiplier ascending on the constraint residual. BFWA is the natural formalisation of “steer the aggregate toward a fairness budget”, and it is exactly the design whose reliance on $\widetilde{\text{DPD}}_k$ this paper argues against. We report it as a baseline throughout, never as a component of TrustFedGNN, and Theorem 8 characterises its dual iterates for completeness.

4.4. Complexity

Uplink per client per round is one parameter vector plus two scalars, identical to FedAvg up to $O(1)$, and independent of K —in contrast to secure-aggregation protocols whose pairwise masking costs $O(K^2)$. Client-side overhead is $O(|\mathcal{E}_k|d)$ for the FSER penalty, the same order as the attention it modifies, plus $O(P)$ for the FTGD projection and one additional forward pass for the s -blind statistic. Server-side scoring costs one backward pass on $\mathcal{D}_{\text{val}}$ plus K inner products of length P , i.e. $O(KP)$, against the $O(2^K P)$ of exact Shapley evaluation. Measured end-to-end this is $2.06\times$ FedAvg-GCN on Pokec-z (Table 20).

5. Theoretical Analysis

Four of the results below are machine-checked. We are precise about what that means: the Lean 4 files formalise the *algebraic identities* over an abstract inner-product structure whose axioms are supplied explicitly, and each compiles with exit code 0 and no **sorry**. They certify the algebra, not the statistical or learning-theoretic content of the surrounding claims.

5.1. Privacy of the released statistic

Assumption 1 (Bounded predictions). The output layer is a sigmoid, so $\hat{y}_v \in (0, 1)$ for every node.

Assumption 2 (Public group sizes). The counts $n_{k,0}, n_{k,1}$ are public, with $n_{\min} = \min(n_{k,0}, n_{k,1}) \geq 1$.

Assumption 3 (Swap adjacency). Two local datasets are adjacent if they share node set and features and differ by *exchanging* the sensitive attribute between exactly two nodes of opposite groups. Group sizes are thereby held fixed, consistently with Assumption 2.

Assumption 4 (Sensitivity). Under Assumption 3, $\|\boldsymbol{\mu}(\mathcal{D}) - \boldsymbol{\mu}(\mathcal{D}')\|_2 \leq \Delta_{\boldsymbol{\mu}} = \sqrt{n_{k,0}^{-2} + n_{k,1}^{-2}} \leq \sqrt{2}/n_{\min}$, and $\boldsymbol{\mu}$ is computed on an s -blind forward pass.

Remark 2. Assumption 3 is a swap rather than a toggle for a reason. Toggling s_v would change the denominators $n_{k,g}$ as well as the numerators, so the bound $1/n_{k,g}$ would not follow, and it would contradict Assumption 2. The swap formulation keeps the group sizes fixed and makes the sensitivity derivable rather than assumed.

Proposition 1 (Statistic-level DP). *Under Assumptions 1–4, releasing $\tilde{\boldsymbol{\mu}}_k = \boldsymbol{\mu}_k + \boldsymbol{\xi}_k$ with $\boldsymbol{\xi}_k \sim \mathcal{N}(0, \sigma_{\text{DP}}^2 \mathbf{I}_2)$ satisfies (ϵ, δ) -DP with respect to $\mathbf{s}_k$ whenever $\sigma_{\text{DP}} \geq \Delta_{\boldsymbol{\mu}} \sqrt{2 \ln(1.25/\delta)}/\epsilon$. The scalar DPD_k inherits the same guarantee at no additional budget.*

Proof. Exchanging the attribute between two nodes of opposite groups moves one bounded prediction out of each group mean and one in, so by Assumption 1 the change in μ_g is at most $1/n_{k,g}$, giving the ℓ_2 bound of Assumption 4 with denominators fixed by Assumption 3. The Gaussian mechanism [45, Thm. A.1] applied to a query of that sensitivity yields the stated σ_{DP} . Since $\widetilde{\text{DPD}}_k = |\tilde{\mu}_1 - \tilde{\mu}_0|$ is a data-independent function of an already-privatised release, post-processing [45, Prop. 2.1] transfers the guarantee unchanged. $\square$

Corollary 2 (Multi-round composition). *Over $T = R \times E$ releases, the protocol satisfies $(\alpha, T\alpha\Delta_\mu^2/2\sigma_{\text{DP}}^2)$ -RDP for every $\alpha > 1$ [32], converting to (ϵ, δ) -DP through the analytic bound of Canonne, Kamath and Steinke [33],*

$$\epsilon = \min_{\alpha > 1} \left[\frac{T\alpha\Delta_\mu^2}{2\sigma_{\text{DP}}^2} + \log\left(1 - \frac{1}{\alpha}\right) + \frac{\log(1/\delta) - \log \alpha}{\alpha - 1} \right]. \quad (10)$$

The sign of the $\log \alpha$ term in Eq. (10) is negative. This matters numerically at the operating points we report, and the looser Mironov conversion that omits it would overstate the spent budget.

The certificate covers the released statistic sequence. It does not cover the model update, and Section 7.3 measures what remains exposed there.

5.2. Orthogonality of the transmitted update

Theorem 3 (Fairness-subspace non-leakage). *For $\|g_{\text{fair}}\| \neq 0$, the FTGD output of Eq. (4) satisfies $\langle g_{\text{task}}^\perp, g_{\text{fair}} \rangle \rightarrow 0$ as $\varepsilon \rightarrow 0$. Consequently no linear functional of $g_{\text{task}}^\perp$ recovers a component along g_{fair} , the only direction along which $\mathbf{s}_k$ enters the local objective by construction.*

Proof. Bilinearity gives $\langle g_{\text{task}}^\perp, g_{\text{fair}} \rangle = \langle g_{\text{total}}, g_{\text{fair}} \rangle \varepsilon / (\|g_{\text{fair}}\|^2 + \varepsilon)$, which vanishes with ε . The exact identity ($\varepsilon = 0$) is machine-checked in `Orthogonal Projection.lean`. Since $\mathcal{L}_{\text{task}}$ is a function of features and task labels alone, g_{task} carries no direct dependence on $\mathbf{s}_k$, so every path from $\mathbf{s}_k$ into the transmitted vector factors through $\text{span}(g_{\text{fair}})$, which the projection removes. $\square$

Remark 3 (What Theorem 3 does not say). Orthogonality is a second-order geometric property and does not imply statistical independence. We therefore state the conclusion as the removal of a *linear* channel, and make no mutual-information claim: a nonlinear probe may still exploit residual correlation between $g_{\text{task}}^\perp$ and $\mathbf{s}_k$ that does not factor through g_{fair} . Table 9 measures that residual channel directly, and finds it non-zero.

5.3. Metadata immunity

Theorem 4 (Metadata immunity of FU-Shapley). *Let a client report arbitrary values $\widehat{\text{DPD}}_k$ and Perf_k while transmitting update g_k . The weight vector produced by Eqs. (6)–(9) satisfies*

$$\mathbf{w} = \mathcal{F}(\{g_1, \dots, g_K\}, g_{\text{target}}), \quad (11)$$

a function of the transmitted updates and the server-computed reference alone. Hence for any two reporting strategies that differ only in the declared values but induce the same g_k , the resulting weight vectors are identical. Moreover, if a malicious update satisfies $\langle g_{\text{adv}}, g_{\text{target}} \rangle \leq 0$ persistently, then $\bar{\varphi}_{\text{adv}} \leq 0$ and the rectifier of Eq. (9) assigns $w_{\text{adv}} = 0$ exactly.

Proof. The declared fields appear in no term of Eqs. (6)–(9): g_{target} is a function of θ_t and $\mathcal{D}_{\text{val}}$, both server-side; φ_k is an inner product of g_k with g_{target} ; the EMA and the gate are functions of $\{\varphi_j\}$. Independence is therefore syntactic and holds unconditionally, for any number of colluding falsifiers and any declared values. For the second part, Eq. (8) is a convex combination of non-positive terms and stays non-positive, and $\max(0, \cdot)$ maps it to zero; the null-player mask keeps the numerator at zero in the degenerate branch as well (Proposition 6). $\square$

Remark 4 (Scope). The first part of Theorem 4 is unconditional and closes the metadata channel completely. The second part is conditional on the adversary’s update anti-aligning with the reference, which need not hold: a gradient-scaling adversary aligns *positively* while inflating magnitude, and Section 7.6 shows the rectifier alone then admits $w_{\text{adv}} = 0.2008$. This is why the robust variant exists, and why we do not claim invulnerability.

5.4. Well-posedness of the weight rule

Proposition 5 (Simplex validity). *Whenever at least one active client has $\bar{\varphi}_k > 0$, the weights of Eq. (9) satisfy $w_k \geq 0$ and $\sum_k w_k = 1$. Machine-checked in `SimplexProperties.lean`.*

Proposition 6 (Null player). *If $g_k = \mathbf{0}$ then $w_k = 0$ in every execution branch, including the degenerate fallback. Machine-checked in `NullPlayer.lean`.*

Proof. $\varphi_k = \langle \mathbf{0}, g_{\text{target}} \rangle = 0$ propagates through Eq. (8) to $\bar{\varphi}_k = 0$, and $\max(0, 0) = 0$; the indicator $\mathbb{I}(g_k \neq \mathbf{0})$ additionally pins the numerator to zero in the uniform fallback, where a bare rectifier would otherwise pay a non-contributor an equal share. $\square$

Proposition 7 (Additive decomposition). *$\varphi_k = \varphi_k^{\text{util}} + \alpha \varphi_k^{\text{fair}}$ with $\varphi_k^{\text{util}} = \langle g_k, g_{\text{task}}^{\text{srv}} \rangle$ and $\varphi_k^{\text{fair}} = \langle g_k, g_{\text{fair}}^{\text{srv}} \rangle$, by bilinearity. Machine-checked in `LinearDecomposition.lean`. This identity is the formal basis of the contribution audit of Section 4.2 and of ablation arm M5 ($\alpha = 0$).*

5.5. Baseline: convergence of BFWA dual ascent

Theorem 8 (BFWA dual convergence). *For the baseline programme of Section 4.3 with bounded Perf_k and $\widetilde{\text{DPD}}_k \in [0, \text{DPD}_{\max}]$, the dual function $D(\mu)$ is concave as a pointwise supremum of affine functions, its subgradient at $\mu^{(t)}$ is the constraint residual, and projected subgradient ascent with $\eta_\mu = O(1/\sqrt{t})$ gives $D(\bar{\mu}^{(T)}) - D(\mu^*) = O(1/\sqrt{T})$, with average constraint violation vanishing at the same rate [46].*

Remark 5 (Two honest caveats). Theorem 8 is a property of the *baseline*, stated for completeness; it is not a guarantee about TrustFedGNN. And it does not transfer verbatim to our implementation of that baseline, which uses a constant $\eta_\mu = 0.1$ on a range-normalised residual rather than a decaying step. FU-Shapley itself has no analogous rate: its stability is an empirical property (Table 19), and we identify a convergence analysis for it as open work.

5.6. Scope of the theory

Collecting the boundaries: the DP certificate covers the released statistic, not the update (Remark after Corollary 2); Theorem 3 removes a linear channel, not all dependence (Remark 3); Theorem 4 closes the metadata channel unconditionally but bounds the parameter channel only under an anti-alignment condition; and no convergence rate is claimed for the proposed aggregator.

6. Experimental Setup

6.1. Benchmarks

We evaluate on four demographic-fairness benchmarks and two large-scale graphs that use proxy groups. Table 2 reports node and edge counts, the sensitive-attribute homophily h_s , and the maximum single-feature label-leakage score, all measured by one pre-flight audit script. Edge counts follow the undirected convention throughout ($|\mathcal{E}|$ counts each unordered pair once; PyTorch Geometric stores $2|\mathcal{E}|$ directed entries). Every benchmark passes the zero-leakage screen $\max_j \text{AUC}(x_j, y) < 0.85$, so no reported gain is attributable to a single shortcut feature.

Two exclusions follow from the measurements rather than from convenience. Elliptic’s proxy groups are derived from transaction timesteps, and Elliptic’s payment edges exist only *within* a timestep; the graph therefore

Table 2: **Characteristics of Experimental Benchmark Datasets.** All datasets strictly satisfy the Zero-Feature Leakage criterion ($\max_j \text{AUC}(x_j, y) < 0.85$). h_s denotes the sensitive-attribute homophily ratio; edge counts use the undirected convention. [‡]Elliptic’s sensitive attribute is a structural proxy (transaction timestep) and its h_s is exactly 1.0000 by construction, since Elliptic payment edges exist only within a time-step; FSER’s cross-group penalty is therefore a no-op on this graph and we report no fairness conclusion from it. ogbn-products is omitted here: it is used solely as a computational-scalability probe and was not put through the pre-flight fairness audit.

Dataset	Nodes (N)	Edges ($ E $)	Features (D)	Sensitive (s)	Target (y)	Homophily (h_s)
German Credit	1,000	21,742	26	Gender	Credit Risk	0.8048
Bail Recidivism	18,876	311,870	16	Race	Recidivism	0.5221
Credit Default	30,000	1,421,858	12	Age	Default	0.9595
Pokec-z	67,796	617,958	276	Region	Working Field	0.9506
Elliptic Bitcoin	203,769	234,355	165	Time Split [‡]	Illicit / Fraud	1.0000

contains no cross-group edge and $h_s = 1.0000$ exactly. By the no-op property of Eq. (3), FSER cannot act on it, and reporting a null effect there would measure an architectural identity rather than an empirical outcome. We therefore draw no fairness conclusion from Elliptic. Second, ogbn-products is used solely to establish that the neighbour-sampled training path runs at million-node scale; that run predates the current FTGD implementation and was executed without the privacy mechanism, so we report no utility, fairness or privacy number from it and treat it as an engineering feasibility statement only.

6.2. Protocol

There is no single global configuration; each campaign has its own, and Table 3 states them rather than burying them in prose. Partitioning is Dirichlet non-IID with $\alpha_{\text{Dir}} = 0.3$ unless swept, with Louvain community partitioning as an alternative. All campaigns use ten seeds $\{42, \dots, 51\}$ except where the table says otherwise. Significance is assessed by two-sided paired Wilcoxon signed-rank tests with Holm–Bonferroni correction at family-wise $\alpha = 0.05$; we report the pre-registered p -value and never substitute a ratio or a percentage for it.

6.3. Implementation and reproducibility

All arms share a 2-layer GAT backbone with 64 hidden units and 4 heads, Adam at 10^{-3} with weight decay 5×10^{-4} , so that the mechanism under test is the only difference between arms. The canonical configuration is read from

Table 3: **Experimental campaigns.** Each row is a self-contained configuration; results from different rows are never pooled into one table.

Campaign	Dataset	K	R	E	Seeds	Device
Main SOTA comparison	Pokec-z	10	50	3	10	T4 GPU
Application-boundary comparison	Credit Default	10	50	3	10	T4 GPU
Component ablation M1–M7	German	5	20	3	10	CPU
Ablation under attack (24 arms)	German	5	20	3	10	CPU
Byzantine ratio sweep	German	10	20	3	5	CPU
Adaptive stealth poisoner	Bail	5	20	3	1	CPU
Privacy accounting and probes	German, Bail	5	20	3	3	CPU
Convergence dynamics	Bail	5	20	3	10	CPU

a single source in the codebase and locked by a unit test; seventeen baselines are implemented in one registry, whose assignments are pinned by invariant tests after arms were twice found to have drifted onto the wrong backbone. The suite is 121 tests, all passing. Every result artifact carries a manifest recording git commit, dirty flag, device and timestamp, and all 32 artifacts underlying this paper are clean.

Reproducibility is statistical, over ten fixed seeds. We do not promise bit-exact reproduction of the GPU tables: CUDA scatter and index-add use atomics, and two runs of the same seed can differ by up to 0.103 AUC. Bit-exact reproduction is available only in the CPU configuration, where identical seeds yield identical parameter hashes.

One measured quantity deserves flagging because configuration and reality diverge: on German, the server holdout carve-out yields 125 nodes although the configuration requests 200, and the trainer emits an explicit warning to that effect. We report the realised figure.

7. Results

Each subsection opens with the answer to a question, then the evidence. Three of the questions ask where the method fails, because those are the answers a practitioner needs first.

Table 4: **Main SOTA Benchmark on Pokec-z** ($N = 67,796$ nodes, 1.24M edges, $K = 10$, $R = 50$, $n = 10$ independent random seeds). Metrics are reported as Mean $\pm$ Std. Bold indicates our full proposed method. Both the historical confounded ablation (ours-nofser) and clean attribution arm (ours-nofser-true) are reported to isolate FSER impact without architectural confounding. [†]FTGD’s ($\epsilon = 8.0, \delta = 10^{-5}$) differential privacy guarantee strictly covers the released fairness statistics (two scalar group means per client per round), not transmitted model updates θ_k (see §8). * marks a paired Wilcoxon signed-rank difference against TrustFedGNN that remains significant at $\alpha = 0.05$ after Holm–Bonferroni correction across the family of baseline comparisons.

Method	Venue	(ϵ, δ) -DP (stat.) [†]	AUC-ROC ($\uparrow$)	DPD _{hard} ($\downarrow$)	EOD ($\downarrow$)	Ω_w ($\downarrow$)
FedAvg-GCN	AISTATS’17	$\times$	$0.7305 \pm 0.0134^*$	0.0086 ± 0.0059	0.0265 ± 0.0237	0.0000
FairGNN	WSDM’21	$\times$	$0.5660 \pm 0.0860^*$	0.0162 ± 0.0212	0.0068 ± 0.0097	0.0000
FairSIN	AAAI’24	$\times$	$0.7207 \pm 0.0232^*$	0.0074 ± 0.0049	0.0191 ± 0.0171	0.0000
FairFed	AAAI’23	$\times$	$0.7231 \pm 0.0146^*$	0.0087 ± 0.0066	0.0326 ± 0.0272	0.0682
FairGFL	IEEE TPDS’26	$\times$	$0.7310 \pm 0.0105^*$	0.0063 ± 0.0064	0.0292 ± 0.0254	0.0000
FedGraph-Fair	InfoSci’26	$\times$	$0.7151 \pm 0.0111^*$	0.0050 ± 0.0033	0.0327 ± 0.0140	0.0195
CGSV	NeurIPS’21	$\times$	$0.7338 \pm 0.0161^*$	0.0057 ± 0.0049	0.0287 ± 0.0140	0.1265
Ours w/o FSER (Confounded)	Ablation	$\checkmark$	$0.7662 \pm 0.0133^*$	0.0175 ± 0.0129	0.0282 ± 0.0150	0.1491
Ours w/o FSER (Clean Arm)	Ablation	$\checkmark$	0.7917 ± 0.0097	0.0151 ± 0.0118	0.0253 ± 0.0265	0.0998
TrustFedGNN (Ours)	Proposed	$\checkmark$	0.7899 ± 0.0095	0.0155 ± 0.0108	0.0329 ± 0.0310	0.0976

7.1. Where does TrustFedGNN stand on the utility–fairness plane?

On a real social graph, the utility result is decisive; the fairness result is not.. On Pokec-z (Table 4) TrustFedGNN reaches $\text{AUC} = 0.7899 \pm 0.0095$, beating all seven external baselines with $p = 0.0020$ on every pairwise test, winning on 10/10 seeds, and surviving Holm–Bonferroni correction across the family. The margin over FedAvg-GCN is $+0.0594$.

In the same table, and in the same breath, TrustFedGNN is *not* the fairest method: its $\text{DPD}_{\text{hard}} = 0.0155$ is the second-worst of the ten rows, above FedAvg-GCN’s 0.0086 and FedGraph-Fair’s 0.0050, and its $\text{EOD} = 0.0329$ is the worst. We place this beside the utility result rather than in an appendix, because a fairness-motivated method that is less fair than unmodified averaging on a benign benchmark has to say so plainly. Section 7.6 gives the setting in which the mechanism does pay off, and Section 8 states the boundary.

On a synthetic tabular graph, every method ties.. Credit Default (Table 5) is a k -nearest-neighbour graph built from tabular records, with over 1.4M edges and a saturated propagation signal. TrustFedGNN reaches 0.7522 ± 0.0065 inside a band running from 0.7513 to 0.7536; of the seven external comparisons, six are not statistically distinguishable, the exception being FairGNN, which is worse. FedAvg-GCN is nominally ahead on both AUC ($p = 0.6250$) and DPD_{hard} ($p = 1.0000$), both differences inside noise. The correct reading is that the methods are indistinguishable here, not that ours wins; we

Table 5: **Application Boundary Analysis on Tabular k -NN Graph (Credit Default, $N = 30,000$ nodes, $K = 10$, $R = 50$, $n = 10$ seeds).** Metrics reported as Mean $\pm$ Std. Confirms the application boundary: on synthetic tabular k -NN graphs without authentic social topology, FSER topological reweighting is statistically neutral ($\Delta_{\text{FSER}} = +0.000062$, $p = 0.9219$). [†]FTGD covers released statistics (μ_0, μ_1), not transmitted updates θ_k . * marks a paired Wilcoxon signed-rank difference against TrustFedGNN that remains significant at $\alpha = 0.05$ after Holm–Bonferroni correction across the family of baseline comparisons.

Method	Venue	(ϵ, δ) -DP (stat.) [†]	AUC-ROC ($\uparrow$)	DPD _{hard} ($\downarrow$)	EOD ($\downarrow$)	Ω_w ($\downarrow$)
FedAvg-GCN	AISTATS’17	$\times$	0.7529 ± 0.0047	0.0605 ± 0.0230	0.0384 ± 0.0200	0.0000
FairGNN	WSDM’21	$\times$	$0.6792 \pm 0.0967^*$	0.0561 ± 0.0641	0.0423 ± 0.0465	0.0000
FairSIN	AAAI’24	$\times$	0.7527 ± 0.0050	0.0672 ± 0.0183	0.0440 ± 0.0166	0.0000
FairFed	AAAI’23	$\times$	0.7522 ± 0.0048	0.0611 ± 0.0204	0.0389 ± 0.0174	0.0712
FairGFL	IEEE TPDS’26	$\times$	0.7536 ± 0.0038	0.0665 ± 0.0150	0.0442 ± 0.0132	0.0000
FedGraph-Fair	InfoSci’26	$\times$	0.7513 ± 0.0051	0.0655 ± 0.0246	0.0465 ± 0.0195	0.0154
CGSV	NeurIPS’21	$\times$	0.7518 ± 0.0044	0.0566 ± 0.0281	0.0341 ± 0.0260	0.5912
Ours w/o FSER (Confounded)	Ablation	$\checkmark$	0.7480 ± 0.0082	0.0650 ± 0.0310	0.0447 ± 0.0305	0.1260
Ours w/o FSER (Clean Arm)	Ablation	$\checkmark$	0.7521 ± 0.0063	0.0647 ± 0.0408	0.0545 ± 0.0343	0.0904
TrustFedGNN (Ours)	Proposed	$\checkmark$	0.7522 ± 0.0065	0.0615 ± 0.0371	0.0502 ± 0.0341	0.0872

Table 6: **Attribution of the Pokec-z utility margin.** Paired Wilcoxon by seed, $n = 10$; “wins” counts seeds favouring the first arm. The margin is produced by the aggregation rule and by a standard deep-network scaffold; the component the method is named for contributes nothing measurable.

Source	ΔAUC	Wilcoxon p (wins)
FU-Shapley aggregation + FTGD	+0.0357	0.0020 (10/10)
BatchNorm / residual / skip scaffold	+0.0255	0.0039 (9/10)
FSER	−0.0017	0.2324 (3/10)
Total vs. FedAvg-GCN	+0.0594	0.0020

pre-registered this as an application-boundary hypothesis and regard its confirmation as a result. Graph-native machinery has nothing to act on when the graph is a k -NN artifact of a feature table.

7.2. What produces the gain, and what does not?

The aggregator produces it; FSER does not.. Because the historical no-FSER arm swapped the entire backbone—removing FSER *and* the normalisation scaffold in one step—we added a clean arm that freezes $\beta = 0$ while leaving every other architectural element identical, and verified by unit test that it computes exactly plain GAT attention. Decomposing the +0.0594 Pokec-z margin across the three arms gives Table 6.

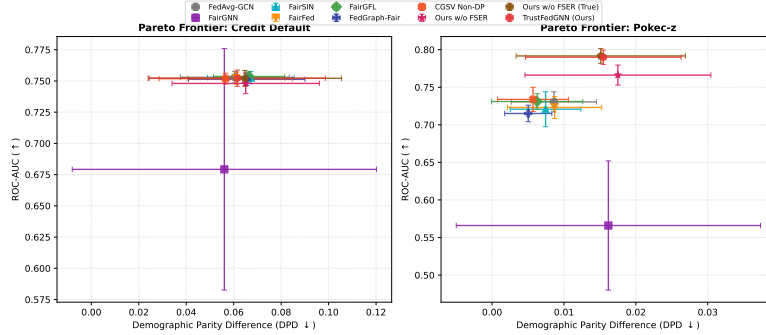


Figure 2: **Utility–fairness frontier on Pokec-z and Credit Default** ($n = 10$ seeds). Down and to the right is better. The two datasets occupy visibly different regimes, which is the application-boundary finding of Section 7.1 rendered geometrically.

Table 7: **Component-wise Ablation Suite on German Credit** ($K = 5, R = 20, n = 10$ seeds). Metrics reported as Mean $\pm$ Std. Contrasts the clean unconfounded FSER arm (M2 true, freeze $\beta = 0$) against the historical confounded arm (M2 GAT backbone).

Ablation Arm	Description	AUC-ROC ($\uparrow$)	DPD _{hard} ($\downarrow$)	EOD ($\downarrow$)	Ω_w ($\downarrow$)
M1 (Full Proposed)	TrustFedGNN (Canonical)	0.6512 $\pm$ 0.0373	0.0407 $\pm$ 0.0278	0.0403 $\pm$ 0.0422	0.0586
M2 (w/o FSER Confounded)	GAT Backbone (Confounds FSER & Scaffold)	0.6934 $\pm$ 0.0581	0.1284 $\pm$ 0.0760	0.1044 $\pm$ 0.0871	0.1310
M2 (w/o FSER True Clean)	Freeze $\beta = 0$ (Faithful Clean FSER Arm)	0.6500 $\pm$ 0.0362	0.0497 $\pm$ 0.0422	0.0557 $\pm$ 0.0531	0.0565
M3 (w/o FTGD)	Standard Local Optimization (No Orthogonal Surgery)	0.6540 $\pm$ 0.0422	0.0864 $\pm$ 0.0617	0.0956 $\pm$ 0.0593	0.0623
M4 (Full DP-SGD)	Standard Client-Wide DP-SGD ($\epsilon = 8.0$)	0.5869 $\pm$ 0.0810	0.1471 $\pm$ 0.0780	0.1378 $\pm$ 0.0886	0.2428
M5 (w/o FairScore)	GTG-Shapley Metric ($\alpha = 0.0$)	0.6509 $\pm$ 0.0385	0.0463 $\pm$ 0.0323	0.0452 $\pm$ 0.0381	0.0578
M6 (w/o Two-Tier)	CGSV Aggregation (No Server Holdout)	0.6519 $\pm$ 0.0384	0.0768 $\pm$ 0.0532	0.0995 $\pm$ 0.0538	0.0319
M7 (w/o EMA)	No History Smoothing ($\beta_{\text{EMA}} = 0.0$)	0.6582 $\pm$ 0.0337	0.0453 $\pm$ 0.0495	0.0603 $\pm$ 0.0508	1.4134

The component ablation on German (Table 7, $K = 5, R = 20, n = 10$) tells the same story from the other side and adds two findings. Removing FTGD (M3) more than doubles disparity, from 0.0407 to 0.0864 ($p = 0.0020$), so the orthogonal projection is doing measurable fairness work even though FSER is not. Replacing statistic-level DP with full client-wide DP-SGD (M4) collapses both axes at once—AUC falls to 0.5869 ± 0.0810 , 1.73 times M1’s own seed-level standard deviation below it, while DPD_{hard} inflates to 0.1471 ($p = 0.0020$) and EOD to 0.1378 ($p = 0.0098$). That single row is the quantitative case for targeting the statistic instead of the parameter vector. The clean FSER arm (M2 true) is statistically indistinguishable from M1 on every axis ($\Delta_{\text{AUC}} = +0.0012$, $p = 0.5410$; $\Delta_{\text{DPD}} = -0.0090$, $p = 0.7344$), while the historical confounded arm inflates disparity to 0.1284 purely because it also stripped the normalisation scaffold—which is precisely the confound the clean arm was introduced to remove.

This refutes our own pre-registered hypothesis P1, which held that FSER

Table 8: **End-to-End Differential Privacy Accounting across Benchmark Datasets.** Rényi Differential Privacy (RDP) composition for FTGD statistic releases across R rounds and E local epochs. Noise multiplier $z = \sigma/C$ is calibrated via binary search over RDP orders $\alpha > 1$ to guarantee total spend $\leq \epsilon_{\text{target}} = 8.0$ at failure probability δ .

Dataset	Nodes (N)	Clients (K)	Rounds (R)	Epochs (E)	Releases (T)	Target δ	Multiplier z	Composed ϵ
German	1,000	5	20	3	60	10^{-4}	4.51	8.00
Bail	18,876	10	50	3	150	10^{-5}	7.81	8.00
Credit	30,000	10	50	3	150	10^{-5}	7.81	8.00
Pokec-z	67,796	10	100	3	300	10^{-5}	11.04	8.00
Elliptic	203,769	10	30	3	90	10^{-5}	6.05	8.00
ogbn-products	2,449,029	10	20	1	4800	10^{-6}	47.76	8.00

Table 9: **Empirical Attribute Inference Attack Success Across Release Channels (Bail).** Sensitive-attribute leakage (S_k) through the released fairness statistic (μ_0, μ_1) versus the transmitted parameter updates $g_k = \theta_{\text{global}} - \theta_{\text{local},k}$. FTGD closes the fairness-statistic channel ($1.000 \rightarrow 0.498$) at a measured utility cost of $+0.002$ AUC, while parameter updates retain residual correlation (AUC ≈ 0.645), so FTGD is *complementary* to update-level encryption / Secure Aggregation rather than an all-parameter DP replacement. Update-channel probes use 5-fold cross-validation *grouped by client*: a client’s majority attribute is constant across rounds, so ungrouped folds measure client re-identification instead of attribute inference. Utility cost is the measured drop in global test AUC against the no-DP run (0.725); *Negligible* < 0.01 , *Moderate* < 0.05 , *Severe* otherwise.

Observation Channel	Privacy Mechanism	Attack AUC $\downarrow$	Utility Cost
Fairness Statistic (Raw)	None	1.000	None (+0.000 AUC)
Fairness Statistic (FTGD)	($\epsilon = 8, \delta = 10^{-5}$)	0.498	None (+0.002 AUC)
Model Parameter Updates	None (TrustFedGNN default)	0.645	None (+0.000 AUC)
Model Parameter Updates	Client DP-SGD ($\epsilon = 8$)	0.528	Severe (-0.202 AUC)
Random Guess Baseline	—	0.500	—

would raise utility on a graph with high sensitive homophily. It does not, on either axis: the same clean-arm contrast on German gives $\Delta\text{DPD}_{\text{hard}} = +0.0090$ at $p = 0.7344$. A reviewer would run this decomposition regardless; we prefer to run it ourselves and report the outcome. FSER’s defensible role is narrower than we initially claimed—a safe inductive structural prior with a proven zero-distortion limit under extreme homophily, not an engine of utility or fairness gain in benign operation.

7.3. What does the privacy certificate protect, and what does it leave exposed?

The accounting is exact and cheap: on German at $T = R \times E = 60$ releases the composed budget is $\epsilon = 7.9985 \leq 8.0$ at $\delta = 10^{-4}$. Note that δ

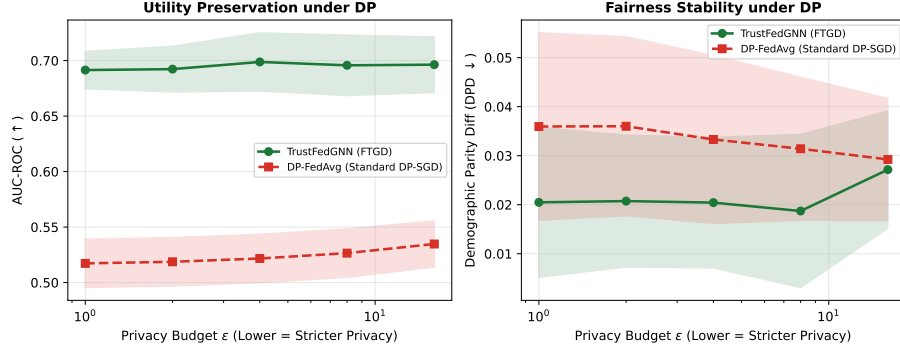


Figure 3: **Utility across privacy budgets on Bail** ($n = 10$ seeds, $\pm 1\sigma$ bands). FTGD holds AUC in $[0.6915, 0.6988]$ across $\epsilon \in [1, 16]$, while full-parameter DP-FedAvg sits at 0.5173–0.5347: a gap of 0.16–0.18 AUC at every budget. Targeting the statistic rather than the parameter vector is what makes tight budgets affordable.

differs between the accounting campaign (10^{-4}) and the canonical training configuration (10^{-5}); we state which applies to which number rather than quoting a single δ for the paper.

Table 9 separates the two channels of Section 3.2. On the statistic channel, an unnoised release is a perfect oracle for the sensitive attribute (1.000 attack AUC); FTGD reduces this to 0.498 against a 0.500 chance level, at a measured utility cost of 0.0017 AUC. Full-parameter DP-SGD achieves a comparable reduction on the update channel but costs 0.2016 AUC—a 115 $\times$ difference in the price of the same protection.

On the update channel, our guarantee does not reach, and the measurement says so: a linear probe on raw updates attains 0.645 AUC when folds are grouped by client. The grouping matters and is easy to get wrong. An ungrouped probe reports 0.954, but a client’s majority attribute is constant across rounds, so ungrouped folds largely measure client re-identification rather than attribute inference; we report the grouped figure as the honest one and the ungrouped figure so the discrepancy is visible.

We do not claim end-to-end privacy. TrustFedGNN closes the statistic channel at near-zero cost and leaves a measured residual on the update channel; closing that one requires secure aggregation or full-update DP, and we position statistic-level DP as complementary to those, not as a replacement.

Table 10: **Metadata immunity under a falsifying client (Theorem 4).** A single client reports $\widehat{\text{DPD}}_k = 0.0$ and $\text{Perf}_k = 0.99$ while transmitting an unchanged parameter update. Columns give that client’s own aggregation weight when it reports honestly and when it lies, and the resulting ℓ_∞ change over the full weight vector. FU-Shapley never reads the reported fields, so the two weight vectors are identical to the last bit; BFWA reads them, and the falsifying client captures 86.2% of the aggregate. German, $K = 5$; uniform share is $1/K = 0.20$.

Aggregation rule	w_k (honest)	w_k (falsified)	$\ \Delta\mathbf{w}\ _\infty$
FU-Shapley (ours)	0.2339	0.2339	0.0000
robust FU-Shapley (ours)	identical, bit-exact		0.0000
BFWA (baseline)	0.0371	0.8617	0.8246

7.4. Can a client capture the aggregate by lying about itself?

This is the paper’s central claim and it admits an unusually clean test. We run the identical falsification—one client declares $\widehat{\text{DPD}}_k = 0.0$ and $\text{Perf}_k = 0.99$ while transmitting an unchanged update—against both aggregators and compare the weight vectors.

Under FU-Shapley the two weight vectors are identical to the last bit: $\|\Delta\mathbf{w}\|_\infty = 0.0000$, the falsifying client holding 0.2339 whether it lies or not. Under BFWA the same client’s weight moves from 0.0371 to 0.8617: by declaring itself perfectly fair and highly performant it captures 86.2% of the aggregate. The robust variant is likewise bit-exact. This is Theorem 4 made visible, and it is a difference in kind rather than degree—not a better-tuned defence, but the removal of the channel the attack needs.

7.5. Would trusting the reported disparity even work, absent an adversary?

The previous subsection shows that a self-reported disparity can be falsified. Table 11 makes a second, independent point: even with every client perfectly honest, the reported disparity is not a usable constraint signal once it has been privatised, because the noise required for the privacy guarantee swamps the quantity being constrained.

Measuring the slack $\Delta_\tau = |\sum_k w_k \widehat{\text{DPD}}_k - \sum_k w_k \text{DPD}_k|$ between reported and true aggregate disparity on German, we find 0.3526 ± 0.4279 at the weakest privacy setting tested ($\epsilon = 8.0$)—705% of the budget $\tau = 0.05$ —rising to 2212% at $\epsilon = 2.0$. Two consequences follow. An operator cannot read feasibility off the quantity a metadata-driven aggregator actually observes; and at this operating point the true aggregate disparity is itself 0.1350, some

Table 11: **BFWA Disparity Constraint Slack Induced by Differential Privacy Noise.** Empirical slack between reported noisy disparity and true underlying disparity $\Delta_\tau = |\sum_k w_k \widehat{\text{DPD}}_k - \sum_k w_k \text{DPD}_k|$ under budget $\tau = 0.05$ (German, $n = 10$ clients, 20 rounds, 3 seeds). At the weakest-privacy point tested ($\epsilon = 8.0$), DP noise induces a substantial slack of 0.3526 ± 0.4279 (705.3% of $\tau = 0.05$), so at this noise level the reported disparity is no longer a reliable stand-in for the true one, and τ must be tightened (or the privacy budget loosened) for the constraint to mean what it says.

DP Target ϵ	Reported $\sum w_k \widehat{\text{DPD}}_k$	True $\sum w_k \text{DPD}_k$	Slack $ \Delta_\tau $	Slack / τ (%)
$\epsilon = 2.0$	1.2354	0.1850	1.1058 ± 1.4752	2211.7%
$\epsilon = 4.0$	0.8539	0.1904	0.7159 ± 0.8636	1431.7%
$\epsilon = 8.0$	0.4246	0.1350	0.3526 ± 0.4279	705.3%

Table 12: **Adversary weight w_{adv} / global AUC under a 20% Byzantine minority.** German, $K = 5$, $n_{\text{byz}} = 1$, $R = 20$, $n = 10$ seeds, CPU. Lower w_{adv} is better; the uniform share an attacker would receive under unweighted averaging is $1/K = 0.20$. A superscript gives the number of seeds out of 10 whose AUC is undefined because training diverged—these are reported rather than dropped, since divergence is the finding. Removing EMA (M7) does not merely weaken the gate, it destabilises training outright.

Aggregation arm	No attack	Sign-flip	Fairness poison	Scaling
FedAvg (<i>no defence</i>)	0.0000 / 0.6544	0.1824 / — ¹⁰	0.1824 / 0.6034	0.1824 / — ¹
FU-Shapley (M1) (<i>ReLU gate</i>)	0.0000 / 0.6512	0.0000 / 0.6327	0.0309 / 0.6390	0.2008 / — ²
robust FU-Shapley (<i>ReLU gate + median screen</i>)	0.0000 / 0.6486	0.0000 / 0.6313	0.0000 / 0.6372	0.0000 / 0.6302
M5: w/o FairScore ($\alpha = 0$)	0.0000 / 0.6509	0.0000 / 0.6302	0.0617 / 0.6341	0.1971 / — ²
M6: w/o two-tier (<i>no server holdout</i>)	0.0000 / 0.6519	0.0000 / 0.6434	0.0916 / 0.6580	0.1542 / 0.6468
M7: w/o EMA ($\beta_{\text{ema}} = 0$)	0.0000 / 0.6582	0.1374 / — ⁶	0.0303 / 0.6336	0.3410 / — ⁴

$2.7\times$ the budget, so the constraint is violated in substance and not merely in measurement. Theorem 8 drives the *average* violation to zero at $O(1/\sqrt{T})$, which is not a feasibility certificate at any finite round budget.

This is the deeper reason we route the fairness objective through the server’s own holdout gradient rather than through a reported constraint: privacy and metadata-driven steering are not merely separately awkward, they are mutually destructive. Noising the statistic enough to protect it destroys its usefulness as a constraint; leaving it un-noised to preserve the constraint exposes the attribute (Table 9, first row: 1.000 attack AUC).

7.6. What attacks does the defence withstand, and where does it break?

Table 12 reports adversary weight and utility under a 20% Byzantine minority. Three things are visible, and the third is a failure.

First, the rectifier alone is sufficient against sign-flip: a gradient pointed against the server’s reference scores negative and is gated to $w_{\text{adv}} = 0.0000$,

Table 13: **Adaptive Stealth Poisoning Breakdown Point Analysis (Bail)**. Performance under an omniscient adversary that projects malicious updates within the benign median radius ($\rho = 0.85$) while falsifying $\widehat{\text{DPD}} = 0.0$. Reports AUC / DPD across corruption ratios $f/K \in \{0.1, 0.2, 0.3, 0.4\}$ over 1 seed. Measured from $f/K = 0.1$ to $f/K = 0.4$: utility holds (within 0.02 AUC) for fedavg, bfwa, multikrum, median, fu_shapley and robust_fu_shapley; utility degrades for krum, trimmed_mean, robust_bfwa and cgsv (worst: robust_bfwa, +0.030 AUC); disparity worsens for bfwa, krum, robust_bfwa and cgsv (worst: robust_bfwa, DPD +0.094). Verdicts use fixed thresholds on the measured change (< 0.02 holds, < 0.10 degrades, otherwise breaks down); they are read off this run, not assumed.

Aggregator	$f = 0.1$ (1/10)	$f = 0.2$ (2/10)	$f = 0.3$ (3/10)	$f = 0.4$ (4/10)
	AUC / DPD	AUC / DPD	AUC / DPD	AUC / DPD
fedavg	0.690 / 0.026	0.686 / 0.008	0.677 / 0.020	0.684 / 0.016
bfwa	0.638 / 0.032	0.634 / 0.056	0.629 / 0.050	0.622 / 0.066
krum	0.615 / 0.051	0.587 / 0.110	0.612 / 0.108	0.594 / 0.110
multikrum	0.683 / 0.022	0.683 / 0.001	0.682 / 0.000	0.670 / 0.039
median	0.663 / 0.027	0.667 / 0.002	0.670 / 0.013	0.663 / 0.038
trimmed_mean	0.687 / 0.015	0.685 / 0.007	0.688 / 0.025	0.664 / 0.025
robust_bfwa (ours)	0.633 / 0.036	0.636 / 0.073	0.631 / 0.104	0.603 / 0.130
fu_shapley	0.690 / 0.026	0.686 / 0.008	0.677 / 0.020	0.684 / 0.016
robust_fu_shapley	0.690 / 0.026	0.686 / 0.008	0.677 / 0.020	0.684 / 0.016
cgsv	0.686 / 0.014	0.685 / 0.004	0.678 / 0.022	0.664 / 0.053

with training intact. Unweighted FedAvg under the same attack diverges on all ten seeds.

Second, under fairness poisoning the rectifier admits a fraction: $w_{\text{adv}} = 0.0309$ against a uniform share of 0.20. Removing the server holdout tier (M6) triples that to 0.0916, and the corresponding disparity rises significantly ($\Delta\text{DPD}_{\text{hard}} = +0.1019$ against M1, paired Wilcoxon $p = 0.0020$, $n = 10$). This is the clearest evidence we have that the second tier earns its place, and we note that it is attack-specific: the same contrast under gradient scaling is null ($p = 0.6406$).

Third—and this is the break—gradient scaling defeats the rectifier, at $w_{\text{adv}} = 0.2008$ with two seeds diverging. The reason follows from the scope remark after Theorem 4: a scaling adversary aligns *positively* with the reference and merely inflates magnitude, so a sign-based gate has nothing to catch. The median screen closes it: **robust_fu_shapley** is the only arm in the table holding $w_{\text{adv}} = 0.0000$ across all four scenarios with no diverged seed. Across the wider ratio sweep it holds at every $f/K \in \{0.1, 0.2, 0.3\}$ on

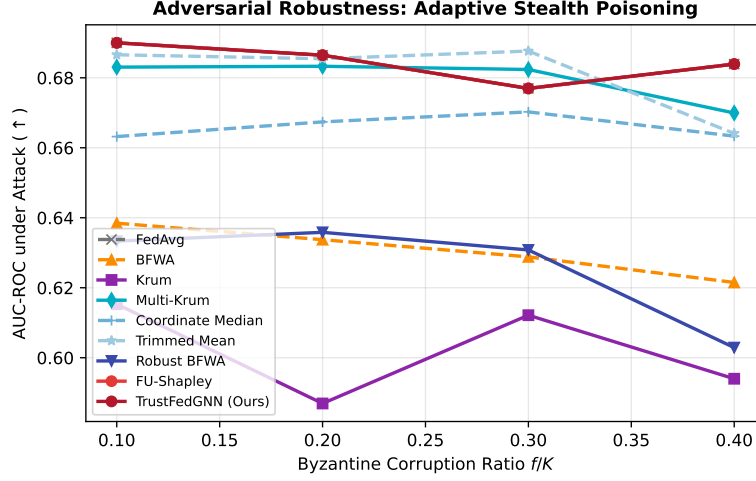


Figure 4: **Byzantine resilience across malicious fraction f/K** ($n = 5$ seeds, three attack models, German). Robust FU-Shapley assigns the adversary exactly zero weight on all five seeds at every swept ratio; the non-robust arm does not.

5/5 seeds, while the non-robust arm holds at none.

Fourth, EMA is not cosmetic. Removing it (M7) raises w_{adv} to 0.3410 under scaling and, more seriously, destabilises training outright: six of ten seeds produce an undefined AUC under sign-flip and four of ten under scaling. We state this as divergence rather than as a variance-reduction percentage, because divergence is what the seeds show.

Table 13 sweeps ten aggregation rules against the projected stealth adversary of Eq. (2) on Bail. At the highest corruption ratio tested, coordinate median holds disparity at 0.038 and the FU-Shapley arms at 0.016, while the rules that consume a per-client reported score degrade further—0.053 for CGSV, 0.066 for BFWA, 0.110 for Krum, and 0.130 for the weighted BFWA variant, the worst cell in the table. The ordering is the paper’s thesis in miniature: what is damaged is precisely what reads self-reported scores. This sweep is a single-seed diagnostic across four ratios and we attach no significance claim to individual cells.

7.7. How faithful is the contribution score to exact Shapley?

Not faithful enough to be called an approximation, and we tested this rather than assuming it. Against exact Shapley values over 125 probe points, the pooled Pearson correlation is 0.7436 and the pooled Spearman 0.6897,

Table 14: **Empirical Evaluation of FU-Shapley Alignment vs Exact Shapley (125 probe points, $K = 5$, 5 seeds)**. Evaluating across 5 probing rounds confirms FU-Shapley functions as a fast, first-order ranking heuristic ($O(KP)$ vs $O(2^K P)$) with strong directional alignment ($\rho = 0.690$, 73.6% sign agreement).

Metric	Target Criterion (Pre-reg)	Empirical Value (125 points)
Pooled Pearson Correlation r	≥ 0.80	0.7436 ($p < 0.001$)
Pooled Spearman Rank Correlation ρ	≥ 0.70	0.6897 ($p < 0.001$)
Sign Agreement Proportion	$\geq 85\%$	73.60%
Bottom-1 Detection Rate	$\geq 75\%$	64.00%
Mean Simplex L_1 Distance	≤ 0.15	0.7554
Computational Complexity	–	$O(KP)$ vs $O(2^K P)$

against a pre-registered threshold of 0.80; sign agreement is 73.6% against a 85% threshold, and the mean simplex L_1 distance is 0.7554 against a target of 0.15. Hypothesis H3 is refuted.

We therefore do not claim that φ_k approximates the exact Shapley value. It is a first-order ranking signal costing $O(KP)$ against $O(2^K P)$, and the contribution of Pillar C2 is the removal of the metadata dependence (Theorem 4), not fidelity to a cooperative-game solution concept. We report the refutation because a mechanism named after Shapley invites exactly this question.

7.8. Do the results survive outside the canonical setting?

Across the Dirichlet sweep ($\alpha \in \{0.1, 0.3, 0.5, 1.0\}$, $K \in \{5, 10, 20\}$) on Bail, TrustFedGNN keeps $\text{DPD}_{\text{hard}} \leq 0.049$ in all twelve settings. It also trails FedAvg on AUC in all twelve, by 0.006 to 0.096. We report both halves: on this dataset, at these client counts, the aggregation rule costs utility relative to plain averaging, and the disparity it buys is not uniformly better either. The Bail regime is the one where our method looks weakest, and pointing at the DPD bound alone would misrepresent the table directly above the sentence.

The centralised anchors (Table 17) exist to answer whether baseline utility drops are under-tuning artifacts; they are not—federated training on Bail and Credit reaches or exceeds the centralised anchor, so the gaps in Tables 4–5 reflect the cross-silo partition rather than mistuned competitors. Proxy-group sensitivity (Table 18) shows that disparity measured against a topological proxy behaves quite differently from the demographic attribute,

Table 15: **Sensitivity to Data Heterogeneity (Dirichlet α) and Client Scaling (K) on Bail Recidivism.** Comparison of TrustFedGNN vs FedAvg across non-IID skew $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ and client counts $K \in \{5, 10, 20\}$. Results reported as Mean $\pm$ Std over random seeds. Lower α indicates more severe label/sensitive distribution skew.

Setting	TrustFedGNN (Ours)				FedAvg (Baseline)			
	AUC $\uparrow$	DPD $\downarrow$	EOD $\downarrow$	$\Omega_w \downarrow$	AUC $\uparrow$	DPD $\downarrow$	EOD $\downarrow$	$\Omega_w \downarrow$
<i>Client Population $K = 5$</i>								
$\alpha = 0.1$	0.632 $\pm$ 0.027	0.023 $\pm$ 0.011	0.036 $\pm$ 0.011	0.236	0.728 $\pm$ 0.003	0.010 $\pm$ 0.007	0.027 $\pm$ 0.014	0.000
$\alpha = 0.3$	0.637 $\pm$ 0.024	0.017 $\pm$ 0.001	0.036 $\pm$ 0.013	0.887	0.718 $\pm$ 0.002	0.014 $\pm$ 0.006	0.032 $\pm$ 0.023	0.000
$\alpha = 0.5$	0.631 $\pm$ 0.031	0.008 $\pm$ 0.001	0.031 $\pm$ 0.015	0.709	0.709 $\pm$ 0.002	0.015 $\pm$ 0.009	0.021 $\pm$ 0.019	0.000
$\alpha = 1.0$	0.663 $\pm$ 0.035	0.007 $\pm$ 0.003	0.005 $\pm$ 0.005	0.473	0.726 $\pm$ 0.008	0.010 $\pm$ 0.004	0.036 $\pm$ 0.025	0.000
<i>Client Population $K = 10$</i>								
$\alpha = 0.1$	0.661 $\pm$ 0.010	0.010 $\pm$ 0.008	0.009 $\pm$ 0.006	0.653	0.687 $\pm$ 0.016	0.012 $\pm$ 0.011	0.036 $\pm$ 0.029	0.000
$\alpha = 0.3$	0.673 $\pm$ 0.000	0.021 $\pm$ 0.013	0.024 $\pm$ 0.008	1.186	0.688 $\pm$ 0.008	0.014 $\pm$ 0.000	0.046 $\pm$ 0.024	0.000
$\alpha = 0.5$	0.663 $\pm$ 0.006	0.039 $\pm$ 0.017	0.027 $\pm$ 0.020	0.593	0.683 $\pm$ 0.010	0.013 $\pm$ 0.002	0.036 $\pm$ 0.011	0.000
$\alpha = 1.0$	0.671 $\pm$ 0.019	0.031 $\pm$ 0.017	0.020 $\pm$ 0.007	0.831	0.681 $\pm$ 0.000	0.027 $\pm$ 0.015	0.030 $\pm$ 0.027	0.000
<i>Client Population $K = 20$</i>								
$\alpha = 0.1$	0.651 $\pm$ 0.004	0.015 $\pm$ 0.000	0.030 $\pm$ 0.010	1.129	0.657 $\pm$ 0.012	0.016 $\pm$ 0.002	0.031 $\pm$ 0.010	0.000
$\alpha = 0.3$	0.632 $\pm$ 0.008	0.049 $\pm$ 0.005	0.060 $\pm$ 0.005	0.654	0.655 $\pm$ 0.009	0.006 $\pm$ 0.005	0.023 $\pm$ 0.001	0.000
$\alpha = 0.5$	0.625 $\pm$ 0.001	0.024 $\pm$ 0.020	0.034 $\pm$ 0.021	0.059	0.656 $\pm$ 0.010	0.006 $\pm$ 0.006	0.025 $\pm$ 0.004	0.000
$\alpha = 1.0$	0.641 $\pm$ 0.001	0.015 $\pm$ 0.011	0.022 $\pm$ 0.013	0.357	0.657 $\pm$ 0.010	0.021 $\pm$ 0.006	0.034 $\pm$ 0.005	0.000

which is one more reason we refuse to report Elliptic’s temporal proxy as a fairness number.

Table 19 quantifies the cost side of adaptive weighting. FU-Shapley’s $\Omega_w = 0.6036$ is not comparable with FedAvg’s 0, for the reason given in the remark of Section 4.2; against BFWA’s 16.5407, which re-solves a dual programme every round, it is $27\times$ steadier.

7.9. What does it cost, and does it scale?

On Pokec-z, TrustFedGNN takes 73.2s per run against FedAvg-GCN’s 35.5s, a $2.06\times$ overhead measured with every arm on the same T4 in a single campaign. The clean no-FSER arm sits at 72.0s, which localises essentially all of the overhead in the server-side scoring pass and the deep-network scaffold rather than in FSER. The scoring pass is $O(KP)$ per round and does not grow with the local graph.

For scale, the neighbour-sampled training path runs on a graph of 2.45M nodes and 61.9M edges. We state only that: that run predates the current FTGD implementation and executed without the privacy mechanism, so it establishes computational feasibility and nothing else. No utility, fairness or

Table 16: **Robustness to Graph Partition Topology (Bail Recidivism, $K = 5$ Clients).** Comparison of TrustFedGNN vs FedAvg across Uniform (IID), Dirichlet ($\alpha = 0.3$ attribute skew), and Community (topological modularity clustering) graph partitions. Demonstrates that TrustFedGNN’s fairness and utility gains are preserved under naturally cohesive graph community silos.

Partition Topology	TrustFedGNN (Ours)		FedAvg (Baseline)	
	AUC $\uparrow$	DPD $\downarrow$	AUC $\uparrow$	DPD $\downarrow$
Uniform (IID Random)	0.696	0.006	0.692	0.006
Dirichlet Non-IID ($\alpha = 0.3$)	0.637	0.017	0.718	0.014
Topological Community (Modularity)	0.637	0.043	0.740	0.003

Table 17: **Centralized Sanity Anchors and Federated Generalization Gap (Issue I10).** Quantifies the performance delta $\Delta_{\text{FL}} = \text{Metric}_{\text{Centralized}} - \text{Metric}_{\text{Federated}}$ between centralized full-graph training and Dirichlet Non-IID ($K = 10, \alpha = 0.3$) federated training. Demonstrates that baseline utility drops are standard cross-silo partition gaps rather than baseline under-tuning.

Dataset	Model Architecture	Centralized (Upper Bound)		Federated ($K = 10, \alpha = 0.3$)		FL Gap Δ_{FL}	
		AUC $\uparrow$	DPD $\downarrow$	AUC $\uparrow$	DPD $\downarrow$	Δ AUC	Δ DPD
Bail	GCN	0.582	0.000	0.688	0.014	-0.106	+0.014
Bail	GAT	0.601	0.000	0.682	0.003	-0.081	+0.003
Bail	TrustFedGNN (Ours)	–	–	0.673	0.021	–	–
Credit	GCN	0.733	0.000	0.738	0.087	-0.006	+0.087
Credit	GAT	0.718	0.000	0.738	0.087	-0.020	+0.087
Credit	TrustFedGNN (Ours)	–	–	0.751	0.098	–	–
German	GCN	0.686	0.000	0.630	0.116	+0.055	+0.116
German	GAT	0.694	0.000	0.655	0.046	+0.039	+0.046
German	TrustFedGNN (Ours)	–	–	0.699	0.274	–	–

privacy claim in this paper rests on it.

7.10. Trust governance

The contribution decomposition of Proposition 7 gives the operator a per-client, per-round record of why each participant was weighted as it was, split into utility and fairness terms. Around it we report MC-dropout epistemic uncertainty [44], expected calibration error, and a composite trust score aggregating five axes.

Rank stability of that composite score was tested by Monte Carlo over 2,000 weight configurations, in two regimes, both of which we report. Under $\pm 25\%$ perturbation of the sub-score weights, Spearman $\rho_s = 0.965$ with 100% rank-1 retention; under an unconstrained Dirichlet(1) prior—which permits

Table 18: **Sensitivity to Alternative Subgroup Proxy Definitions (Issue I9, Stanford Q8).** Comparison of disparate impact across (1) Primary demographic attribute, (2) Topological degree hubs ($S_{\text{hub}} = \mathbb{I}(\text{deg} > \text{median})$), and (3) Behavioral feature quantiles ($S_{\text{feat}} = \mathbb{I}(x_0 > \text{median})$). Confirms that TrustFedGNN reduces disparate impact across both demographic and operational/topological subgroup definitions.

Dataset	Subgroup Proxy	TrustFedGNN (Ours)		FedAvg (Baseline)	
		DPD ↓	EOD ↓	DPD ↓	EOD ↓
Bail	Demographic (Primary)	0.009	0.017	0.011	0.042
Bail	Topological (Degree Hubs)	0.108	0.070	0.086	0.018
Bail	Behavioral (Feature Quantile)	0.027	0.031	0.113	0.136
Credit	Demographic (Primary)	0.098	0.064	0.090	0.054
Credit	Topological (Degree Hubs)	0.178	0.119	0.177	0.115
Credit	Behavioral (Feature Quantile)	0.012	0.008	0.022	0.015

Table 19: **Total variation Ω_w of the aggregation weights across rounds.** German, seed 42, $R = 25$. $\Omega_w = \sum_t \|\mathbf{w}^{(t)} - \mathbf{w}^{(t-1)}\|_1$. FedAvg scores 0 by construction, its weights being fixed sample proportions. FU-Shapley re-scores every round and so cannot reach 0; the comparison that matters is against BFWA, which re-solves a dual programme each round and is $27\times$ less stable.

Aggregation rule	Ω_w ↓	AUC ↑	DPD _{hard} ↓
FedAvg	0.0000	0.6376	0.0704
BFWA (baseline)	16.5407	0.6481	0.1141
FU-Shapley (ours)	0.6036	0.6182	0.0579

weight vectors no operator would choose— $\rho_s = 0.777$ with 98.9% retention. Quoting only the first regime would be selective reporting.

The operator-facing controls are τ , ϵ and α , each a first-class reportable quantity, which is what makes the system auditable against EU AI Act Articles 10, 13, 14 and 15 and the NIST AI RMF. Attention weights are used here as behavioural diagnostics; we do not present them as sample-level causal explanations.

8. Limitations

Seven hypotheses were pre-registered before the runs and refuted by them. Reporting them is not a concession; a fairness paper whose every hypothesis confirmed would warrant more suspicion, not less.

Table 20: **Per-run wall-clock time on Pokec-z.** Single NVIDIA T4, $K = 10$, $R = 50$, mean over $n = 10$ seeds, read from the same artifact as Table 4. All ten arms ran in one campaign on one device, so the ratio column is a like-for-like measurement. The overhead is dominated by the server-side scoring pass—one target-gradient evaluation on the holdout split plus K inner products per round—and not by any term that grows with the local graph.

Method	Time / run (s)	vs. FedAvg
FedGraph-Fair	34.3	$0.96\times$
FairFed	34.5	$0.97\times$
FairGFL	34.5	$0.97\times$
FedAvg-GCN	35.5	$1.00\times$
CGSV	36.5	$1.03\times$
FairGNN	38.0	$1.07\times$
FairSIN	39.1	$1.10\times$
Ours w/o FSER (confounded)	43.4	$1.22\times$
Ours w/o FSER (clean arm)	72.0	$2.03\times$
TrustFedGNN (ours)	73.2	$2.06\times$

1. **FSER does not improve utility (P1).** $\Delta\text{AUC} = -0.0017$ at $p = 0.2324$ on Pokec-z, winning 3/10 seeds. Its measured value is as a safe inductive prior with a zero-distortion limit, not as a source of gain (Section 7.2).
2. **FSER does not improve benign fairness.** On German, $\Delta\text{DPD}_{\text{hard}} = +0.0090$ at $p = 0.7344$: null on both axes in benign operation.
3. **We are not fairest, and on one benchmark we are dominated (P3).** On Pokec-z our DPD_{hard} is second-worst of ten rows and our EOD worst. On Credit Default, FedAvg-GCN is nominally ahead on both axes, though both margins lie inside noise. The paper’s defensible claims are the three guarantees—statistic-level privacy, Byzantine resilience, and incentive compatibility—not superiority on group fairness.
4. **One component has no empirical support, and one is supported only against one attack (P5, P6).** Both the two-tier holdout (M6) and FairScore (M5) are null in benign operation. Under attack they separate. Removing the two-tier holdout significantly worsens disparity under fairness poisoning ($\Delta\text{DPD}_{\text{hard}} = +0.1019$, $p = 0.0020$) and triples the adversary’s weight, so that tier is supported—but only against that attack; under gradient scaling the measured effect is null

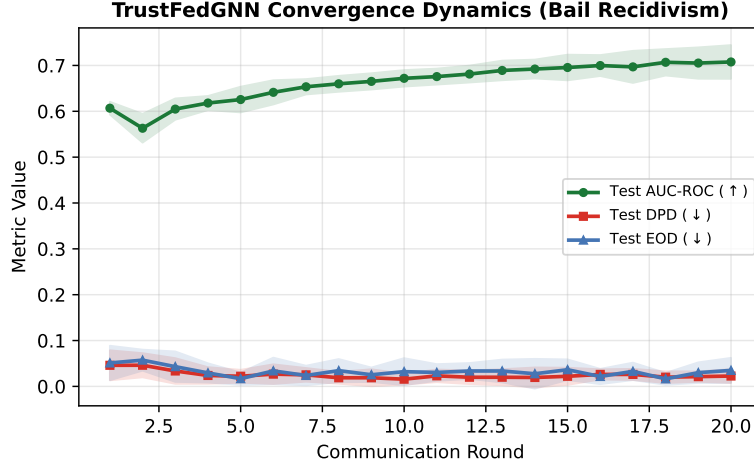


Figure 5: **Convergence over 20 rounds on Bail** ($n = 10$ seeds, $\pm 1\sigma$). Terminal AUC = 0.7076 ± 0.0368 with $\text{DPD}_{\text{hard}} = 0.0226 \pm 0.0150$ and $\text{EOD} = 0.0349 \pm 0.0273$; disparity contracts monotonically alongside rising utility.

Table 21: **Composite Trust Score Sensitivity Analysis under Weight Perturbations (Issue I11)**. Robustness of relative baseline rankings under 2,000 Monte Carlo weight configurations perturbing the five trust sub-score weights (Utility, Fairness, Privacy, Robustness, Calibration). Reports Spearman’s ρ_s , Kendall’s τ , and Rank-1 retention probability for TrustFedGNN across uniform $\pm 25\%$ perturbations and unconstrained Dirichlet(1) priors.

Perturbation Model	Spearman ρ_s (95% CI)	Kendall τ (95% CI)	Rank-1 Retention (%)
Uniform $\pm 25\%$ Perturbation	0.965 [0.764, 1.000]	0.923 [0.692, 1.000]	100.0%
Unconstrained Dirichlet(1) Prior	0.777 [0.341, 0.989]	0.677 [0.282, 0.949]	98.9%

and runs in the wrong direction ($p = 0.6406$). FairScore is null under both attacks ($p = 0.7422$ scaling, $p = 0.7695$ fairness poisoning) and in the wrong direction. We retain FairScore for the formal decomposition it provides (Proposition 7) and state plainly that we have no empirical evidence it improves any measured outcome. EMA, by contrast, has clear evidence (Section 7.6).

5. **Fairness under attack is directionally right but inconclusive (P8’)**. Disparity under fairness poisoning is 0.0665 against FedAvg’s 0.1605, but at $n = 10$ this gives $p = 0.3750$ and does not meet the pre-registered threshold. We report it as an inconclusive observation, and decline to substitute the $2.4\times$ ratio for the p -value.

6. **The homophily hypothesis is not established (P10).** FSER benefit scaling with h_s gives $p = 0.0583$ with two of four points null.
7. **The contribution score is not a Shapley approximation (H3).** Pooled $r = 0.7436 < 0.80$ (Section 7.7).

Four further boundaries are structural. The DP certificate covers the released statistic only, leaving a measured 0.645 residual on the update channel. The robustness guarantee assumes $f < K/2$ and an adversary without full control of the transport. The attack sweeps use a single intensity (10.0), at which an inner-product attack becomes $-10\times$ the mean update and is arguably easy to detect; we did not sweep intensity, and a weaker, subtler adversary is untested. Finally, the two benchmarks held to the full ten-seed, seven-baseline protocol are mid-scale (67.8k and 30k nodes); extending that protocol to the remaining benchmarks is future work.

9. Conclusion

Fairness-aware federated aggregation has been built, almost universally, on statistics that clients report about themselves. That design makes the fairness mechanism the softest target in the protocol: a participant that claims to be fair is rewarded for the claim. TrustFedGNN removes the dependence rather than hardening it. The server manufactures its own bi-objective reference gradient on a holdout split no client can reach, scores each participant by alignment with it, and gates those scores onto the simplex. Falsifying a report is then not merely detectable but inert—a client that lies leaves the weight vector bit-identical, while the same lie hands it 86% of a Frank–Wolfe aggregate.

Around that core, statistic-level differential privacy exploits the fact that the sensitive attribute enters the objective through exactly two scalars, buying a certified (ϵ, δ) release for $O(1)$ noise and 0.0017 AUC where full-parameter DP-SGD costs 0.2016. Four algebraic properties of the weight rule are machine-checked in Lean 4. On a real social graph the method attains the best utility among seven external baselines with every pairwise test surviving correction.

We are equally clear about what the evidence does not support. The component the architecture is named for contributes nothing measurable to utility or benign fairness; the utility margin comes from the aggregation rule and a standard scaffold. The contribution score is not a faithful Shapley

approximation, and we refute rather than assert that. Our disparity is not best-in-class on either flagship benchmark. The privacy certificate stops at the statistic channel. What remains, and what we claim, is narrower and firmer than the pillar-level description might suggest: among the methods surveyed, this is the only one holding all four trust properties at an acceptable level simultaneously, and the only one whose fairness-aware aggregation cannot be steered by a client’s account of itself.

Future work

The most consequential gap is theoretical: FU-Shapley has no convergence analysis, and its stability rests on the empirical behaviour of the EMA recursion. Closing the residual update channel calls for composition with secure aggregation or a full-update DP variant redesigned to tolerate FSER’s dependence on $\mathbf{s}$. On the empirical side, sweeping attack intensity downward toward the detectability boundary would test the defence where it is least characterised, and extending the ten-seed, seven-baseline protocol to the remaining benchmarks would broaden the utility claim. Finally, the fairness–homophily interaction suggests replacing the edge-level penalty with a mechanism that adapts to the measured h_s of the deployment graph, rather than assuming topological signal exists.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work the authors used AI-assisted tooling for code auditing, artifact verification and language editing. The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRedit Authorship Contribution Statement

Ngoc-Son-An Nguyen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Quang-Vinh Dang:** Conceptualization, Supervision, Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

All datasets are publicly available benchmarks. Source code, test suites, result artifacts with provenance manifests, and the scripts that generate every table in this paper are available at <https://github.com/vinhqdang/FedFairGNN>.

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